

Standards for Official statistics on Climate-Health interactions

Heat- and cold-related mortality and morbidity topic: background information

1. Acknowledgments

To be added

2. Heat- and cold-related health outcomes and climate change

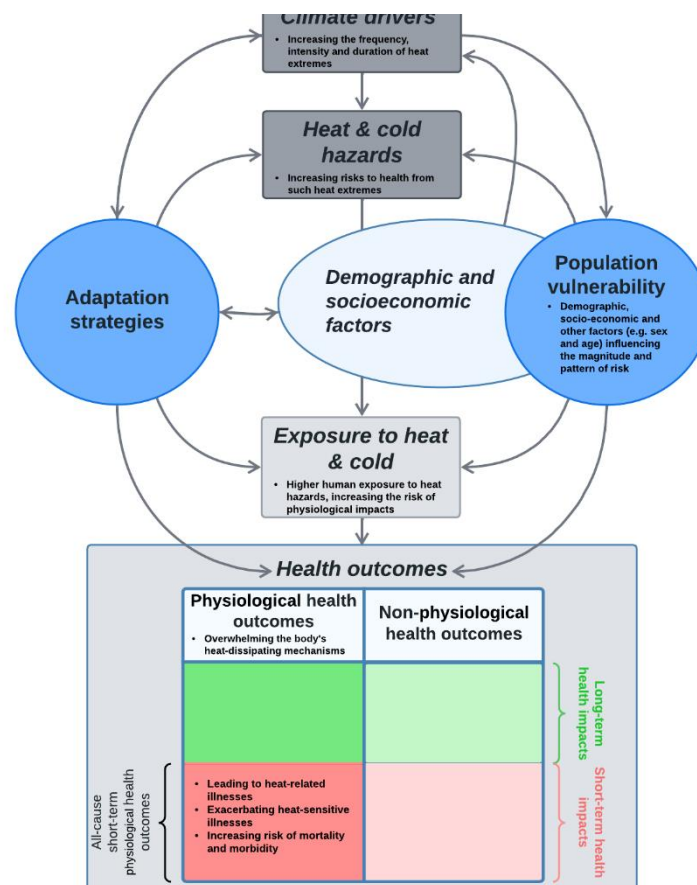
2.1. Definition and scope

Climate change is altering temperature patterns across the world. It is increasing the frequency, intensity, and duration of heat extremes which in turn increase heat-related health risks from human exposure to such heat extremes. Patterns in cold temperatures are consequently also changing as the climate changes. This happens across the globe with research showing an evident attributable increase in mortality due to climate change in cities on every continent (Mitchell, 2021). Changes in such temperature patterns and in the ensuing health outcomes can, thus, indicate how climate change is impacting human health through its impact on temperatures.

Add more on inclusion of topic area and set out key concepts.

2.2. Impact pathway

Figure 1 - Pathways between climate change and heat-health outcomes



Existing studies highlight many complex pathways by which temperature exposure interacts with temperature-sensitive physiological mechanisms that, in turn, affect health. **Error! Reference source not found.** describes a simplified high-level pathway for heat exposure.

Hot environments can overwhelm the body's heat-dissipating mechanisms and redirect blood flows for internal organs. This can lead to negative health impacts or death by causing multiorgan failure (Hifumi, et al., 2018). Cold environments lead to higher blood pressure from narrowing blood vessels along with decreased cardiac output and potential organ failure. Increased instances of influenza are also recorded during periods of low temperature (Kephart, et al., 2022).

3. Health impacts

The health impacts of interest from heat and cold in this topic are the short-term physiological ones. That is, those health outcomes attributable to ambient extreme temperatures with no extra step on the pathway which arise shortly after the exposure. This is because causality between climate change and health outcomes is more easily attributable when investigating the short-term impact of extreme temperatures on health.

From existing research, we know that both mortality and morbidity are affected by extreme heat and cold. Illnesses can be caused as well as exacerbated from extreme temperatures. Some common illnesses caused or exacerbated by heat are heatstroke and other heat illnesses, dehydration, cardiovascular disorders, respiratory disorders, kidney disease, diabetes, mental health conditions, urinary tract infections, adverse birth outcomes, fluid disorders, and sepsis. Cold temperatures are associated with hypothermia, cardiovascular disorders and respiratory disorders including worsening asthma/COPD.

4. Framework indicators

The rate of increase in climate-related mortalities from heat is projected to off-set the rate of decrease in mortalities from cold, leading to net mortality from non-optimal ambient temperatures increasing globally (Zhao, et al., 2021). In order to track the difference in their respective health impacts, the framework contains separate outcomes indicators for heat and for cold.

There are less indicators for cold than for heat since the science relating to cold-related impacts on health from climate change remains less mature than heat and consensus less robust. However, some of the heat indicators, especially where there is a related cold indicator in the UN Global Set, have been complemented with cold-related ones.

4.1. Outcomes indicators

Our framework contains two outcome indicators:

Indicator 1.1: Mortality attributed to non-optimal temperature

Indicator 1.2: Hospitalisations attributed to non-optimal temperature

The indicators are meant to produce daily estimates. However, daily numbers on a localised scale may be too small to adhere to [disclosure standards](#), or daily observations may not be available for every country. Therefore, the indicators are given for an unspecified timescale and framework users decide whether to utilise daily or weekly data.

The indicators cover health outcomes from all causes to ensure that these indicators are applicable in countries where cause-specific outcomes data is not yet available. However, where such data exist, these indicators are broken down by certain leading causes of death/hospitalisation so that the physiological outcomes of non-optimal heat and cold can be identified. These leading causes cover illnesses caused by extreme temperatures as well as exacerbated pre-existing ones which also contribute to the mortality and morbidity burden attributable to extreme temperatures.

To determine whether a day's temperature is considered extreme heat the indicators use hot days, heatwaves, and hot nights as definitions. For extreme cold, the indicators use cold days and cold nights as definitions.

Hot/cold days: Individual days where the temperature is considered non-optimally hot/cold. There are no global standardised definitions for hot/cold days which these indicators draw on. The daily maximum temperature is used for their determination, identifying days where the maximum temperature is above/below a threshold for extreme heat/cold.

Hot/cold nights: Individual nights where the temperature is considered non-optimally hot/cold. There are no global standardised definitions for hot/cold nights which these indicators draw on. The daily minimum temperature is used for their determination, identifying nights where the minimum temperature is above/below a threshold for extreme heat/cold.

Heatwaves: The [UNDRR-ISC Hazard Definition and Classification Review](#) defines a heatwave as “a marked unusual period of hot weather over a region persisting for at least two consecutive days during the hot period of the year based on local climatological conditions, with thermal conditions recorded above given thresholds”. This definition is used to identify heatwaves as at least two consecutive hot days, where hot days are defined as above. This framework does not restrict heatwaves to the hot period of the year to ensure a full representation of the temperature impact on health throughout the year.

Relative rather than absolute temperature thresholds are used to define the hazards. This considers that populations living in countries with more extreme climates tend to be more acclimatised to such temperatures and therefore, require different temperature thresholds for extreme temperature events than others.

Since the health outcome changes with the intensity of the extreme event, two thresholds are chosen for both extreme heat and cold events, to compare the impact of moderate-intensity and of high-intensity extreme temperature events.

4.2. Other indicators

The recommended indicators for other parts of the pathway between climate drivers and health outcomes are:

ID	Topic sub-area	Indicator	Measuring	Related indicators
1.3	Heat & cold	Temperature records – heatwaves, hot/cold days, and hot/cold nights	Hazard	UN global set of climate change statistics and indicators: Temperature records
1.4	Heat & cold	Person-days exposure to temperature records	Exposure	UN global set of climate change statistics and indicators: Temperature records

		- heatwaves, hot/cold days, and hot/cold nights		
1.5	Heat & cold	Exposure of vulnerable populations to the extremes of heat and cold	Exposure & vulnerability	Lancet Countdown 2022: Exposure of Vulnerable Populations to Heatwaves UKHSA Climate change and public health indicators scoping review: Proportion of housing stock with overheating risk
1.6	Heat	Heat information for health (heat surveillance system, early warning system, and health sector response plan)	Adaptation	Lancet Countdown 2022: Climate Information for Health
1.7	Heat & cold	Number of dwellings with adequacy of building materials defined by national or local standards	Adaptation	UN global set of climate change statistics and indicators: Buildings adapted to climate change
1.8	Heat	Share of green urban areas in the total area of cities	Adaptation	UN global set of climate change statistics and indicators: Share of green urban areas in the total area of cities Lancet Countdown 2022: Urban green space

An indicator which combines vulnerability with exposure was chosen for indicator 1.5. This is because vulnerability to temperature extremes is closely linked to a person's exposure to them. For example, poorer populations, urban populations, populations of no fixed abode, outdoor workers, migrant workers, and others experience higher levels of exposure to temperature extremes, which, in turn, contributes to their increased vulnerability. The populations identified as vulnerable are over-65s, children and infants, pregnant women, those with pre-existing health conditions, those living in urban areas, populations living in poverty, populations of no fixed abode, and groups less able to manage their heat and/or cold exposure (e.g. outdoor workers, migrant workers, refugees, army recruits and workers wearing added protective equipment).

Indicators 1.6 to 1.8 were identified since they measure different aspects of adaptation, they have been developed for other (mostly global) frameworks, and can inform the implementation of adaptation strategies. They cover common adaptation measures such as heat action plans (indicators 1.6), the adaptation of the built environment (indicator 1.7) and urban green space (indicator 1.8). Indicator 1.7 also provides an indication of how indoor exposure to temperature extremes may vary over time. If outdoor temperature is extreme but the built environment is well-adapted, the health impact will be milder. This is especially true for cold-related health impacts. However, due to limited data sources on indoor temperature, none of the suggested indicators directly measure indoor temperatures.

5. Further reading

- [Mitchell, D., 2021. Climate attribution of heat mortality. *Nature Climate Change*, pp. 467-468.](#)
- [Zhao, P. Q. et al., 2021. Global, regional, and national burden of mortality associated with non-optimal ambient temperatures from 2000 to 2019: a three-stage modelling study. *Lancet Planetary Health*, 5\(7\), pp. E415-E425.](#)
- [Hifumi, T., Kondo, Y., Shimizu, K. & Miyake, Y., 2018 . Heat stroke. *J Intensive Care*.](#)
- [Kephart, J. L. et al., 2022. City-level impact of extreme temperatures and mortality in Latin America. *Nat Med*, 28.](#)
- [Liu, J. et al., 2022. Heat exposure and cardiovascular health outcomes: a systematic review and meta-analysis. *Lancet Planet Health*, p. 484–95.](#)